

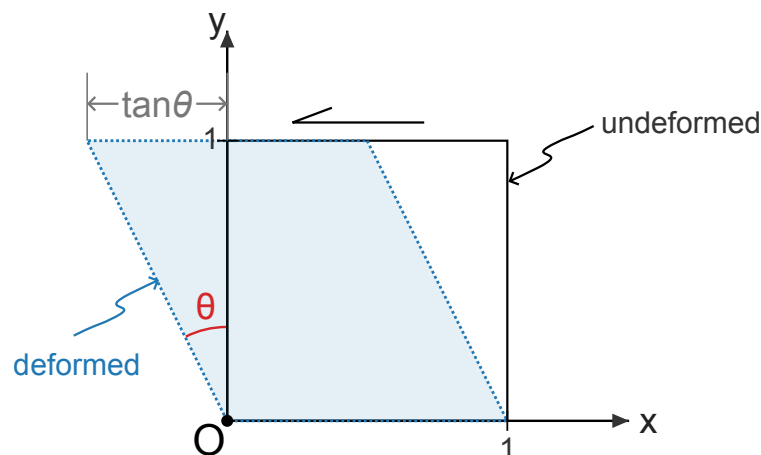
# Finite Strain Analysis and Shear Zones

## Instructions

1. No submission required for Questions 1 and 2. They will be discussed in class.
2. Provide brief answers to Questions 3 and 4. Also print out your **Jupyter Notebook** scripts associated with both question, (with all figure outputs included) for submission.
3. Submit your measurement steps and results for Question 4.

## Question 1 (in class)

We have derived the  $\mathbf{F}$  matrix for the pure shear deformation in class. Now derive the  $\mathbf{F}$  for sinistral simple shear deformation with a shear strain of  $\gamma = \tan \theta$ . Show your reasoning. The following diagram of sinistral simple shear deformation is provided to you as a reference:



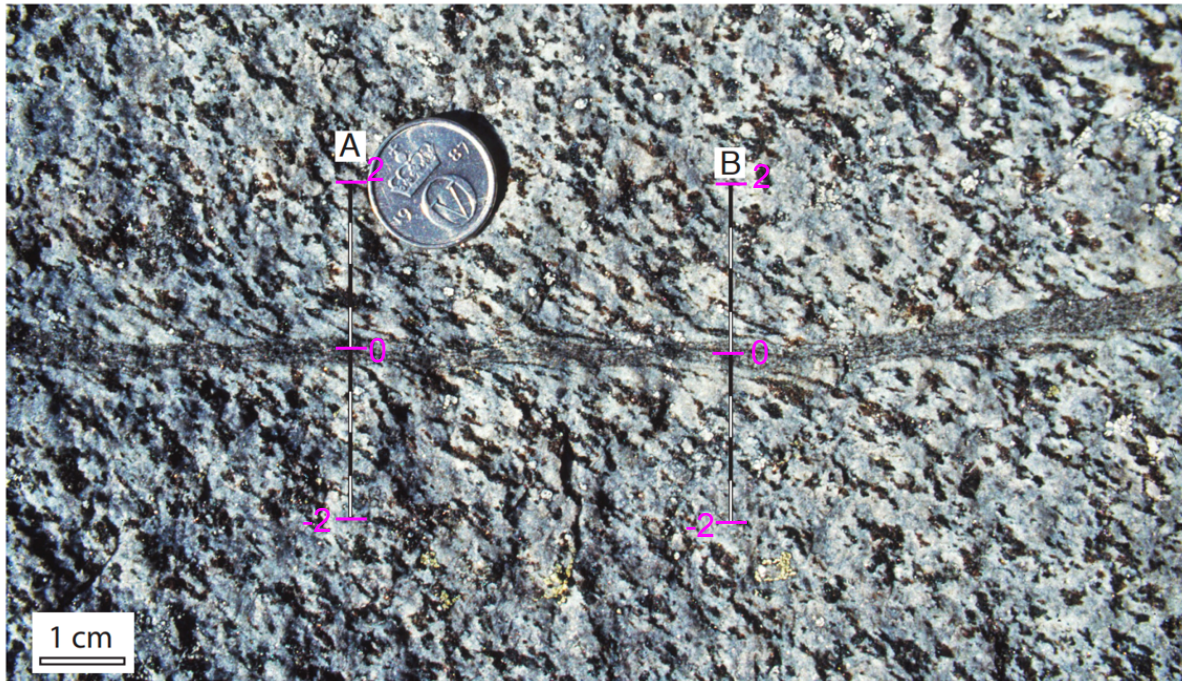
## Question 2 (in class)

Rerun the `finite_strain_analysis.ipynb` with your derived  $\mathbf{F}$  for the sinistral simple shear with increasing shear strain, from  $\theta = 30^\circ$  to  $60^\circ$ . Show figure outputs from the **Jupyter** notebook and briefly explain:

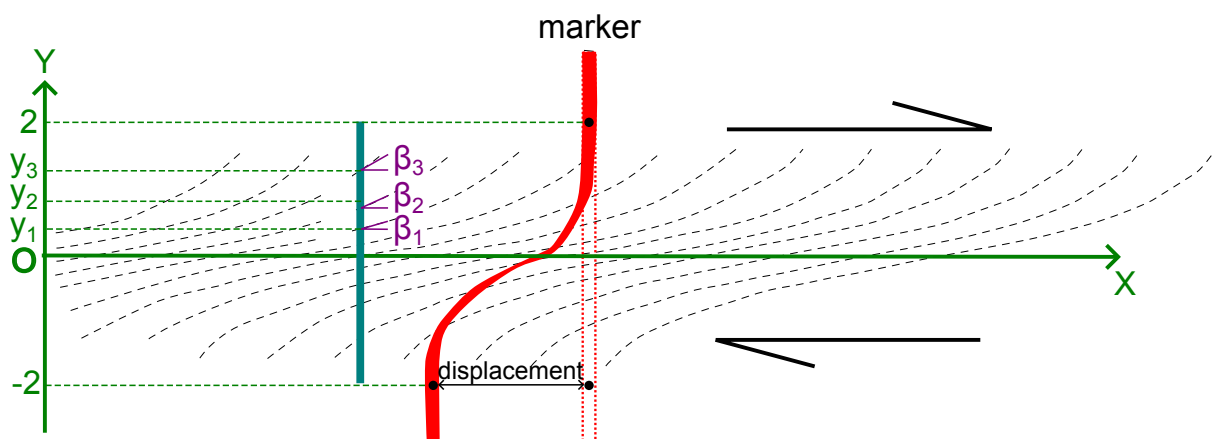
1. Why simple shear deformation is considered as *non-axial*?
2. What is the sense of rotation of the material lines? Are there any materials lines that are not rotated at all?
3. Are material lines experiencing shortened at  $\theta = 30^\circ$  still under shortening as we accumulate shear strain and reach  $\theta = 60^\circ$ ?

### Question 3

The following figure displays a narrow shear zone developed in a magmatic rock. Now let's do some "forward modelling" on a virtual piece of rock sample. Plug the  $\mathbf{F}$  you computed in Question 1 into `homogeneous_strain.ipynb`. Experiment with different shear strain ( $\gamma$ ) values, do the modelled strain ellipsoids produce the similar deformation pattern as the natural shear zone in the figure?



### Question 4



1. Create two shear strain profiles, A and B, across the shear zone. On the transparent paper, mark at least 8 oblique fabrics along each profile and measure the angles ( $\beta$ ) between the foliation and the shear zone boundary. Record the position and  $\beta$  angle of the measurements.

2. Average the measurements along both profiles and input the values into the notebook file `heterogeneous_strain.ipynb` to construct a spatially varying shear strain  $\gamma$  profile. The  $\beta$  values has the following relationship with the shear strain  $\gamma$ :  $\gamma = 2/\tan 2\beta$  and is already implemented for you in the notebook.
3. Compute the overall displacement  $d$  (cm) of the shear zone along each profile using the following relationship:  $d = \int_{-2}^2 \gamma d\gamma$ .  
(Hint: consider using the function `scipy.integrate.simpson` )
4. Show the modelled deformation pattern from the figure. Is it more realistic than the previous model? What does that tell you about the strain distribution within the shear zone?